

30062 GENERAL MATHEMATICS (BIEM-BIEF) DETAILED SYLLABUS – A.Y. 2025/2026

First part

Structures

[Sets, operations with sets.] *Property of the double complement (H). De Morgan's laws (*)*. [Number sets. The set $\mathbf{R}$ of real numbers, its geometric representation. Operations. Order. Intervals.] The extended real line $\bar{\mathbf{R}}$. Upper bound, lower bound, maximum, minimum of sets $A \subseteq \mathbf{R}$. Bounded sets $A \subseteq \mathbf{R}$. *Uniqueness theorem for the maximum/minimum (H)*. Supremum, infimum of sets $A \subseteq \mathbf{R}$. Completeness theorem for $\mathbf{R}$ (the supremum principle). Absolute value, distance. Neighborhoods. Left/right neighborhoods. Neighborhoods of $-\infty / +\infty$. *Characterization theorem for supremum/infimum, using left/right neighborhoods (H)*.

The set $\mathbf{R}^n$ of real vectors with n components, its geometric representation for $n = 2, 3$. Linear operations, linear combinations, inner product (or scalar product) of vectors. Order. Positive / strictly positive / strongly positive vectors. Convex linear combinations, segments, convex sets. Norm and its properties; Cauchy-Schwarz inequality, *triangle inequality (H)*. Orthogonal vectors, orthonormal vectors. The Pythagorean theorem in $\mathbf{R}^n$. Distance and its properties. Neighborhoods. Interior, exterior, boundary points. Open, closed, bounded, compact sets. *All neighborhoods are open sets (*)*. *A set is open if and only if its complement set is closed (H)*. Properties of the union and the intersection of open/closed sets. Isolated, limit points. A set is closed if and only if it contains all its limit points. Interior, boundary, closure, derived set of a set.

Compare paragraphs on textbook: § 1.1-1.2; 1.4-1.5; 1.7; 2.1-2.4; 3.3; 4; 5.1-5.6; 8.8 (only 8.8.4); 16.1. Main definitions: 1, 2, 3, 5, 6, complement set, intervals in $\mathbf{R}$, 30, 32, 33, 34, 39, the extended real line, 49, 50, vector, operations with vectors, order with vectors, positive / strictly positive / strongly positive vector, 82, absolute value, norm, 117, 120, orthonormal vectors, 126, 129, 131, 133, exterior point, interior of a set, 136, boundary of a set, 141, 144, derived set, 149, 153, 156, 172, 176, 325, 820. Main results: 8 (H), 16 (*), properties of the order in $\mathbf{R}$, 31, 37 (H), 41, 51, 52, 53, properties of the order with vectors, 113, 114, 115 (H), 119, 127, 132 (H), 152 (*), 158 (H), 164, 170, 171, 823.

Functions

[The concept of function. Domain, codomain, range; image, preimage. Injective, surjective, bijective function. Composition of functions. Inverse function.

Real functions of one real variable; natural domain, graph. Review of elementary functions. Bounded functions. Monotone functions, strictly monotone functions.] Sum and product of functions. *A monotone function is invertible if and only if it is strictly monotone (*)*. Strict monotonicity is equivalent to an order isomorphism. (Global) maxima/minima, maximizers/minimizers; the argmax/argmin notation. (Global) strong maxima/minima, maximizers/minimizers. A maximizer/minimizer is strong if and only if it is unique. Supremum, infimum. Local maxima/minima, maximizers/minimizers. Concave/convex functions, strictly concave/convex functions. Affine functions.

[Real functions of n real variables.] Examples: Cobb-Douglas functions, utility functions, production functions. [Real functions of two real variables: natural domain, graph.] Level curves, indifference curves. Bounded functions. (Global) maxima/minima, maximizers/minimizers. (Global) strong maxima/minima, maximizers/minimizers. Supremum, infimum. *Preservation of maximizers/minimizers with respect to strictly increasing composition (*)*; applications to Economics. Local maxima/minima, maximizers/minimizers. Concave/convex functions, strictly concave/convex functions. Affine functions.

Maximizers (minimizers) of concave (convex) functions. *Fenchel's theorem: local-global maximum (minimum) property for concave (convex) functions (*)*.

Level sets. Upper contour sets, lower contour sets. *Concave (convex) functions have convex upper (lower) contour sets (*)*. Monotone functions on an interval have both convex upper contour sets and convex lower contour sets. Quasi-concave/quasi-convex functions, strictly quasi-concave/quasi-convex functions. *Concave (convex) functions are quasi-concave (quasi-convex) (H)*. Monotone functions on an interval are both quasi-concave and quasi-convex. Characterization of quasi-concave (quasi-convex) functions in terms of convex upper (lower) contour sets. Applications to utility functions; preservation of quasi-concavity with respect to strictly increasing composition.

Compare paragraphs on textbook: § 6; 17 except 17.1.3, 17.2.2, 17.3.2, and 17.3.4; 22.1; 22.5-22.6 except 22.6.2. Main definitions: 183, Cobb-Douglas function, 195, utility function, production function, 196, level curve, level set, indifference curve, 204, product of functions, 208, injective function, surjective function, bijective function, 218, bounded function, 228, 245, 261, 276, 878, 886, 894, 910, 1050, argmax/argmin, 1052, 1122. Main results: 226, 229 (*), 232, 896 (*), 904, 905, concave (convex) functions are quasi-concave (quasi-convex) (H), 911, 912, 913, 918, 1053, 1060 (*), 1067, 1069, 1124 (*).

Sequences

Sequences of real numbers. Recursively defined sequences. Properties that are at least eventually satisfied. Limits of sequences. Convergent, divergent, irregular sequences. Limits from above, limits from below. *For an eventually positive sequence x_n , $x_n \rightarrow +\infty$ if and only if $1/x_n \rightarrow 0$ (*)*. *Theorem on the uniqueness of the limit (* only in the case of finite limits)*. *Boundedness theorem for convergent sequences (H)*. *Regularity theorem for monotone sequences (*)*. Calculation of limits. Limits of elementary sequences. Operations with limits, indeterminate forms. *Limit of a sum of sequences (* only in the case of finite limits)*. *Comparison criterion (*)*. Ratio criterion (only the limit form). Comparisons among infinities and among infinitesimals. The number e . The symbols $\sim$ and o , their properties. *Theorem on the characterization of $x_n \sim y_n$ with the use of o (H)*. Extension to sequences of real vectors.

Compare paragraphs on textbook: § 8.1; 8.5-8.11 except 8.9.2 and 8.11.3; 8.13-8.14 except 8.14.7 and 8.14.8; 8.16. Main definitions: 298, recursively defined sequence, bounded sequence, monotone sequence, 311, 313, 314, 321, 323, 328, 363 only cases (i) and (iii), 386. Main results: 324 (*), 329 (*, only in the case of finite limits), 333 (H), 334 (*), 345 (*, only the sum of sequences, in the case of finite limits), 350 (*), 353 (limit form only), 361, 368, 371 (H), scales of infinities.

Series

The concept of series. The sequence of partial sums. Behavior of a series; convergent, divergent, irregular series. *Behavior of the geometric series (*)*. *Behavior of the Mengoli series (H)*. Behavior of the harmonic series. *Necessary condition for convergence (H)*. Series with non-negative terms: *regularity theorem (H)*, behavior of the generalized harmonic series, asymptotic comparison criterion, comparison criterion. Series with terms of indefinite sign: simple convergence and absolute convergence. The exponential series.

Compare paragraphs on textbook: § 9 except 9.1.2, 9.3.2, 9.3.3, and 9.4.2. Main definitions: 388, 413. Main results: 389 (H), 390, 391 (*), 392 (H), 394 (H), 395, 398, 400, 401, 410, 411, 414, 419.

Limits of functions and continuity

Limits of functions of one real variable (as $x \rightarrow x_0$, as $x \rightarrow +\infty$, as $x \rightarrow -\infty$; from the left, from the right; from above, from below). Vertical and horizontal asymptotes. Characterization of limits with limits of sequences. *Theorem on the uniqueness of the limit (H only in the case of finite limits)*. Calculation of limits. Limits of elementary functions. Operations with limits, indeterminate forms. *Limit of a sum of functions (* only in the case of finite limits)*. *Comparison criterion (H)*. Change of variable. The symbols $\sim$ and o . Fundamental limits. Continuity for functions of one real variable. Characterization of continuity with the ε - δ condition. Characterization of continuity with limits of sequences. Points of discontinuity. Continuity of elementary functions, operations with continuous functions. Weierstrass's theorem, zero-value theorem (Bolzano's theorem) and its application to market equilibria, intermediate-value theorem (Darboux's theorem). Strict monotonicity is equivalent to invertibility, for continuous functions on an interval.

Limits of functions of n real variables. Calculation of limits for functions of two real variables: comparison criterion, change of variable, calculation of limits along different directions. Continuity for functions of n real variables. Weierstrass's theorem; outline on its application to the consumer problem.

Coercive functions; preservation of coercivity with respect to strictly increasing composition for continuous functions, coercivity of continuous functions on compact sets. *Tonelli's theorem (*, excluding the remark on C closed)*. Supercoercive functions; characterization via the upper contour sets, relation between supercoercivity and coercivity.

Compare paragraphs on textbook: § 12; 13.1-13.6 except 13.6.3; 22.1.6, 22.1.7, 22.2-22.3; 36.2. Main definitions: 520, donut-neighborhood, 524, 525, 526, 528, 530, 532, 533, 536, 542, vertical asymptote, horizontal asymptote, 543, 562 only cases (i) and (iii), 570, discontinuities, different kinds of discontinuities, 593, 1090, 1102. Main results: 538, 545, 546, 549 (H, only in the case of finite limits), 551 (H), 554 (*, only the sum of functions, in the case of finite limits), fundamental limits, 567, 577, 578, 587, 588, 590, 592, 595, 596, 600, 602, 1084, 1085, 1093, 1095, 1096, 1100 (*, excluding the remark on C closed), 1103, 1105, 1107.

Second part

One-variable differential calculus

Difference quotient, derivative at a point, derivability; geometric meaning, equation of the tangent line. Left and right derivative at a point; corners, cusps. The derivative function. [Derivatives of elementary functions. Rules on derivatives.] *Derivative of a constant (H)*. *Derivative of $y = x^a$ (H)*. *Derivative of $y = e^x$ (H)*. *Derivative of a linear combination (*)*. Derivative of the inverse function. *Derivative of $y = \ln x$ (H)*. *Derivative of $y = \arctan x$ (H)*. Higher-order derivatives. *Relationship between derivability and continuity (*)*. Differentiability, differential. *Relationship between derivability and differentiability (H)*. Characterization of the possible discontinuities of the derivative function. $C^{(1)}$ functions, $C^{(n)}$ functions, $C^{(\infty)}$ functions.

Continuity, uniform continuity, and the Lipschitz condition. Uniformly continuous functions are continuous; a continuous function defined on a compact set is uniformly continuous. Lipschitz functions are uniformly continuous. Relationship between derivability and the Lipschitz condition.

Stationary points. *Necessary condition for local maximizers/minimizers (Fermat's theorem) (H)*. *Rolle's theorem (H)*. *Lagrange's mean value theorem (H)*. *Characterization of functions with a null derivative (H)*. *Characterization of functions with the same derivative (H)*. *Monotonicity test on an interval (*)*. *Strict monotonicity test on an interval (H)*; its use as invertibility test on an interval. First sufficient condition for local maximizers/minimizers. De L'Hopital's theorem. Taylor's polynomial and Maclaurin's polynomial. Taylor's theorem; Taylor's formula and Maclaurin's formula of order n , with Peano remainder. *Maclaurin's formulae for elementary functions $y = e^x$ (H), $y = \sin x$ (H), $y = \cos x$ (H), $y = \ln(1+x)$ (H), $y = (1+x)^a$* . Second sufficient condition for local maximizers/minimizers. Determination of global maximizers/minimizers. Concavity as a property of tangent lines for differentiable functions. Concavity/strict concavity test on an interval for differentiable functions. Concavity/strict concavity test on an interval for twice-differentiable functions. Necessary and sufficient condition for maximizers (minimizers), for a concave (convex) differentiable function. Study of the graph of a function.

Compare paragraphs on textbook: § 13.9; 19.1; 26.1-26.13 except 26.13.2; 28; 29.1-29.2 except 29.1.3; 31.1 except 31.1.4; 31.5 (only 31.5.1); 36.3. Main definitions: 635, 973, difference quotient, 1315, 1320, 1322, 1347, 1348, $C^{(1)}$ functions, 1354, $C^{(n)}$ functions, $C^{(\infty)}$ functions, 1411, 1463, 1465. Main results: 636, 637, 638, 974, 975, 976, 1317 (H), 1323, 1327 (*), 1328, 1330 (H), 1331, 1332 (*), 1333, 1334, 1336, 1340, 1341 (H), 1342 (H), 1344 (H), 1349 (H), 1350, 1406 (H), 1415 (H), 1417 (H), 1418 (H), 1419 (H), 1423, 1428 (*), 1430 (H), 1435, 1436, 1439, 1442, 1449, 1451, 1466, 1467, 1468 (H), 1469 (H + H + H), 1476, 1539, 1542, 1544, 1589.

Introduction to n -variable differential calculus

Partial derivatives, gradient vector, partial derivability. The derivative operator. Differentiability, differential. Relationship between partial derivability, differentiability and continuity. $C^{(1)}$ functions.

Stationary points. Unconstrained optimization: *necessary condition for unconstrained local maximizers/minimizers (Fermat's theorem) (H)*. Saddle points. Necessary and sufficient condition for maximizers (minimizers), for a concave (convex) differentiable function.

Compare paragraphs on textbook: § 27.1-27.2 except 27.2.3; 27.4; 28.1; 31.5 (only 31.5.1). Main definitions: 1367, 1370, 1373, $C^{(1)}$ functions, saddle point, 1411. Main results: 1369, 1374, 1376, 1377, 1409 (H), 1591.

Linear algebra

Vector spaces. Subspaces; characterization with the closure with respect to linear combinations. *The intersection of subspaces is a subspace (*)*. Linear independence, linear dependence; characterization with linear combinations. Span of a set of vectors; *characterization with linear combinations (*)*. Spanning set. Basis, dimension. *Property of unique writing for a basis (*)*. The standard basis in $\mathbf{R}^n$. Linear independence property of orthogonal vectors. Orthonormal bases in $\mathbf{R}^n$. *Determination of the coefficients of a vector in $\mathbf{R}^n$, with respect to an orthonormal basis (H)*. Linear functions $f: \mathbf{R}^n \rightarrow \mathbf{R}$: definition, *Riesz representation theorem (H)*, continuity of linear functions. Increasing / strictly increasing / strongly increasing real functions of n real variables. Positive / strictly positive / strongly positive real functions of n real variables. Riesz-Markov theorem. Theorem on the characterization of affine functions. Matrices. Linear operations with matrices. Row-column product. Transpose matrix. Power of a square matrix. Symmetrical, triangular, diagonal matrices. Linear operators $T: \mathbf{R}^n \rightarrow \mathbf{R}^m$: definition, *Riesz representation theorem for operators (*)*, representation

matrix, continuity of linear operators. Image space; *the image space is a subspace spanned by the image of the standard basis (*)*. Kernel; *the kernel is a subspace (H)*. Rank and nullity of a linear operator, theorem on rank and nullity; surjectivity, injectivity, bijectivity conditions. Determinant of a square matrix: definition, first Laplace's theorem, properties. Inverse matrix of a square matrix; definition, existence, uniqueness, explicit computation in terms of the adjoint matrix, properties. *Determinant of the inverse matrix (H)*. Rank of a matrix: rank by columns, rank by rows, and relation with the notion of rank of a linear operator. Computation of rank as the maximum order of non-null minors, Kronecker's theorem.

Linear systems: explicit writing, matrix writing, column writing. Existence of solutions: *Kronecker-Capelli's theorem (*)*, distinction between the determined/undetermined cases. Structure of solutions: homogenous systems, non-homogenous systems. Computation of solutions: *Cramer's theorem (H)*, Cramer's rule; Cramer's method for the extension to all linear systems. Introduction to the problem of least squares.

Compare paragraphs on textbook: § 3; 4.2; 6.4 (only 6.4.5); 15.1-15.8 except 15.4.4, 15.6.2, and 15.7.3; 17.1; 22.10 (only 22.10.1).

Main definitions: vector space, 68, 75, 82, span, 93, 95, 102, 107, orthonormal basis, increasing / strictly increasing / strongly increasing real function of n real variables, 673, positive /strictly positive/ strongly positive real function of n real variables, matrix, operations with matrices, symmetric matrix, triangular matrix, diagonal matrix, transpose matrix, 701, image space, kernel, 719, 725, inverse matrix, submatrix, 746, singular matrix, adjoint matrix, 1150. Main results: 69, 73 (*), 84, 87, 88 (*), 94 (*), 99, 101, 103, 106, 121, 123 (H), 236, 675, 677, 679, 683 (H), 685, 686, 687, properties of operations with matrices, 699, 702, 703, 707, 709, 710 (*), 713, 715, 717 (H + *), 718, 720, 722, 723, 727, 729, 738, 744, 747, 748, 749, 753, 756, 757, 758, 759, 760, determinant of the inverse matrix (H), 761, 763, 768, 772, 774, 777 (H), Cramer's rule, 784 (*), 785 (case ii), 786, 787, 788, Cramer's method, 888, 1151, 1152.

Notes

1. All topics appearing in square brackets are taught during the Preparatory Course in Mathematics and are only synthetically resumed in this course.
2. For all topics marked with (*), the proof is given during classes. For all topics marked with (H), the proof is assigned to students as a Homework.

Textbook:

Simone Cerreia-Vioglio, Massimo Marinacci, Elena Vigna, *Principles of Mathematics for Economics*, draft version, August 2025.